

Mike Ly

626-878-4367 | mch3jy@virginia.edu | [LinkedIn](#) | [GitHub](#) | [Personal Website](#)

EDUCATION

University of Virginia

Charlottesville, VA

Bachelor of Arts in Computer Science | GPA: 4.0/4.0 | Dean's List

Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, Computer Systems

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++, SQL, HTML/CSS

Frameworks & Libraries: Node.js, Express.js, React, MongoDB, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib

Testing: pytest, JUnit, Vitest, Playwright

Tools & Databases: Git, GitHub, MongoDB, Jupyter Notebook, VS Code, IntelliJ

Certifications: Google Cybersecurity Professional Certificate

PROJECTS

CommitGate | [GitHub](#)

Jun 2026 – Present

Python, Typer, Git, Gitleaks, OpenAI API, pytest

- Developed a Python CLI that integrates into Git workflow to automatically detect secrets and security vulnerabilities and prevent them from entering version control history
- Constructed a security scan pipeline combining Gitleaks deterministic secret detection with LLM-powered semantic code review for context-based code analysis
- Designed and implemented a unified security finding system that parses Git diffs, normalizes Gitleaks and AI scan results and blocks commits based on configurable security thresholds
- Developed automated tests using pytest to validate Git operations, hook installation and security finding detection

Full-Stack Notes Application | [GitHub](#) | [Live Demo](#)

Dec 2025 – Aug 2026

React, Node.js, Express.js, MongoDB, Mongoose, JWT, Vitest, Supertest, Playwright

- Built and deployed an end-to-end note-taking application with a React frontend, Node.js and Express backend with MongoDB Atlas database using Mongoose to persist user and note data
- Implemented JWT-based authentication, bcrypt password hashing and REST API endpoints for creating, retrieving, updating and deleting notes
- Developed frontend component tests with Vitest and React Testing Library, backend API integration tests with Supertest and end-to-end tests with Playwright

EEG Emotion Recognition using CNN | [Paper](#)

Jul 2024 – Jul 2025

Python, PyTorch, PyTorch Lightning, TorchEEG, scikit-learn

- Designed and implemented a CNN in Python based on TSception architecture with additional temporal kernels and a four-region spatial convolution module
- Built an EEG data processing pipeline for baseline removal, Z-score normalization and five-fold cross-validation
- Trained and evaluated model on DREAMER dataset using PyTorch Lightning
- Achieved 86.58% dominance, 85.24% arousal and 76.87% valence classification accuracy, outperforming the TSception baseline across all three emotion dimensions

EXPERIENCE

Incoming Teaching Assistant, Data Structures & Algorithms

Fall 2026

University of Virginia

Charlottesville, VA

- Selected to support students in Java, object-oriented programming, concurrency, inheritance, polymorphism, data structures and algorithm analysis through office hours, labs and review sessions

Quality Assurance Engineer

Feb 2026 – May 2026

Lawly - Law School Admissions Support

Remote

- Identified, documented, and communicated 80+ software defects across six development sprints by conducting regression, exploratory and end-to-end testing of a Next.js law school admissions support platform
- Validated UI against Figma designs and performed cross-browser and cross-platform testing across Chrome, Edge and Firefox on Windows, macOS, iPhone and iPad to identify layout and functional inconsistencies
- Used Chrome DevTools and Postman to test frontend and API endpoints by analyzing console errors, status codes, request parameters, response payloads and responsive layouts